

Modeling lepton-nucleus scattering cross sections using transfer learning technique

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Electron-nucleus cross sections from transfer learning

Phys. Rev. Lett. 135, 052502

K.Graczyk, B.Kowal et al.

Empirical fits to inclusive electron-carbon scattering data obtained by deep-learning methods

Phys. Rev. C 110, 025501

B.Kowal, K.Graczyk et al.

Inclusive electron-nucleus cross section models from domain adaptation

arxiv.org/abs/2609.08463

K.Graczyk, B.Kowal et al.

- ▶ Motivations
- ▶ Goals
- ▶ Obtaining the base model
 - ▶ Construction of the neural network and the training procedure
 - ▶ Results of the neural network predictions
- ▶ Transfer learning
 - ▶ Layer-wise fine tuning
 - ▶ Results in various kinematic areas



logo of NuWro - a Monte Carlo neutrino event generator

- ▶ The motivations come from studies of **neutrinos'** fundamental properties (UWr Division of Neutrino Physics)
- ▶ Obtaining a **deep learning model of neutrino-nucleus scattering** could help us gain a deeper understanding of neutrino-matter interactions.
- ▶ There are some similarities between electron and neutrino scattering.
- ▶ The first step towards this goal is to create a neural network describing electron-nucleus scattering.

- ▶ The examining that the **Deep Neural Network (DNN)** encodes nuclear structure features
- ▶ The demonstration that the **Transfer learning (TL)** technique can be efficient in modeling physical systems.

The neural network trained on one type of data learns universal properties, that can be transferred do other related problems

- ▶ Obtaining a **electron-nucleus scattering model** using deep learning techniques

$$DNN(E, \theta, \omega) \rightarrow \frac{d^2\sigma}{d\cos(\theta)d\omega} \quad (1)$$

DNN - Deep Neural Network prediction,

E = Energy, *θ* = scattering angle, *ω* =transfer of energy

- ▶ Step 1: Obtaining the **base model** with encoded physics
- ▶ Step 2: Application of **Transfer learning**
- ▶ Development of techniques that allow us estimate the uncertainty of the predictions of DNN.

Step 1

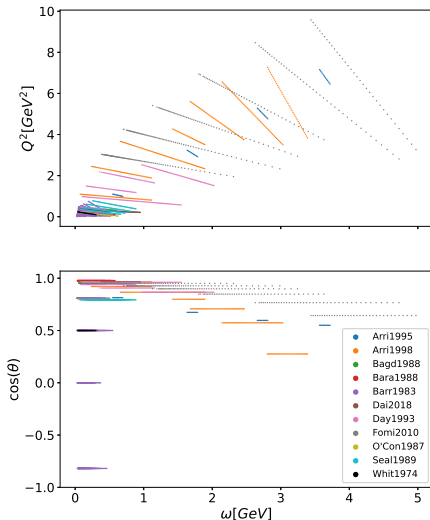
Obtaining the **base model** with encoded physics

We consider the inclusive electron-carbon scattering:

$$e + {}^{12}\text{C} \rightarrow e + X$$

Electron-carbon scattering model

- Data from the [Electron Nucleus Scattering Archive](http://discovery.phys.virginia.edu/research/groups/qes-archive/notes.html):
<http://discovery.phys.virginia.edu/research/groups/qes-archive/notes.html>
- we concentrate on electron-carbon data
- a broad kinematic region: **quasielastic scattering**, **pion production**, and the onset of **deep-inelastic scattering**
- We remove data with the lowest ω by applying an appropriate cut, because they correspond to elastic scattering and discrete nuclear state



- ▶ 11 independent datasets.
- ▶ the k -th dataset containing N_k points
$$\mathcal{D}_k = \{(E_k^i, \theta_k^i, \omega_k^i, d\sigma_k^i, \Delta d\sigma_k^i) : i = 1, \dots, N_k\},$$
where $d\sigma_k^i$ and $\Delta d\sigma_k^i$ are the i -th measurement in k -th dataset and corresponding uncertainty.
- ▶ $\Delta d\sigma_k^i$ includes statistical and systematic uncertainties.
- ▶ The normalization λ , systematic uncertainty, is taken into consideration.

Abbrev.	Number of points
Arri1995	56
Arri1998	398
Bagd1988	125
Bara1988	259
Barr1983	1243
Dai2018	177
Day1993	316
Fomi2010	359
O'Con1987	51
Seal1989	250
Whit1974	31
Total	3265

- ▶ Cross-section values are ineffective, for DNN, range domain
- re-scale the cross-sections (rescaled data < 1)

$$d\sigma \rightarrow \left(\frac{10^9}{137^2 E \cos(\theta/2)} \frac{\cos(\theta/2)^2}{4E^2 \sin(\theta/2)^4} \right)^{-1} d\sigma, \quad (2)$$

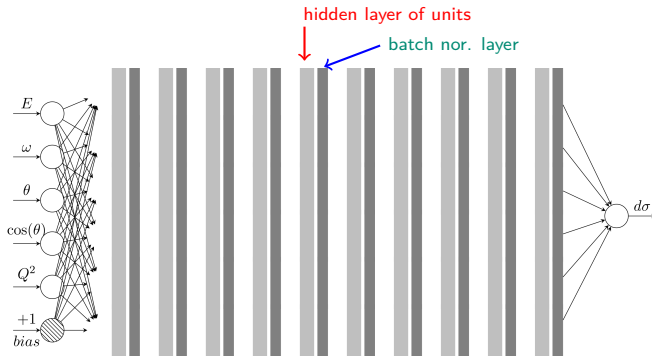
- ▶ Additional input parameters, that depend on the others, to achieve fitting :

$$(E, \omega, \theta) \rightarrow (E, \omega, \theta, \cos(\theta), Q^2)$$

- ▶ DNN may over-fit the data
- ▶ How to estimate the uncertainty of the predictions?

DNN: Model A (Bootstrap model)

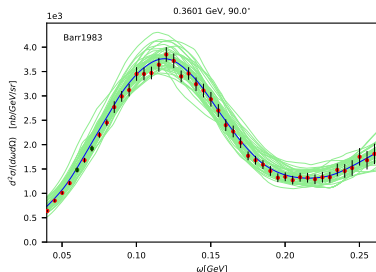
- ▶ 10 blocks with 300 fully connected neurons and following batch normalization layer



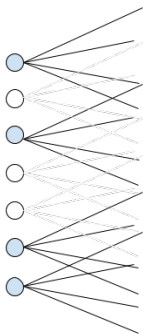
- ▶ Batch Normalization (Ioffe and Szegedy, [arxiv:1502.03167](https://arxiv.org/abs/1502.03167)): re-centering and re-scaling inputs of the layers, improves optimization and regularize the model

Model A: Bootstrap approach

- ▶ Efron (1979): Bootstrap methods
 - ▶ Adapted for neural networks by Tibshirani (1996) and Breiman (1996).
-
- i For each data sample, we have a Gaussian distribution with
 - mean = data point $d\sigma_k^i$
 - std. deviation = uncertainty $\Delta d\sigma_k^i$
 - ii we draw $M = 50$ bootstrap (clone) datasets from Gaussian distributions
 - iii For each bootstrap (clone) data set, obtain DNN fit.
 - iv Averaging over the models prevents overfitting. Get the mean and standard deviation

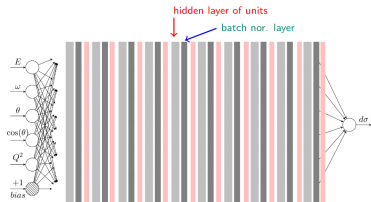


Model B: MC Dropout approach



Dropout layer: In every layer, some hidden units are dropped from the processing the signal (forward and backward), with a probability p

Bootstrap model generalizes better



- ▶ We keep dropout layers active in training mode
- ▶ We obtain one DNN fit

→ To make prediction:

- ▶ We keep dropout layers active in inference mode
 - i compute $M = 50$ times the response of the network for a given input.
 - ii average over the predictions, get the mean and standard deviation

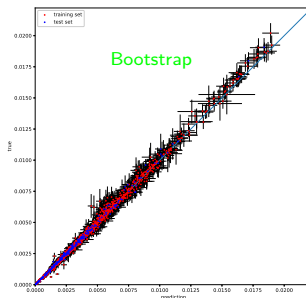
$$\chi_{\text{tot}} = \sum_{k=1}^{11} \left[\chi_k^2(\lambda_k) + \frac{1}{2} \left(\frac{1 - \lambda_k}{\Delta \lambda_k} \right)^2 \right],$$

$$\chi_k^2(\lambda_k) = \frac{1}{2} \sum_{i=1}^{N_k} \left(\frac{d\sigma_k^i - \lambda_k d\sigma_i^{\text{fit}}(E_k^i, \theta_k^i)}{\Delta d\sigma_k^i} \right)^2$$

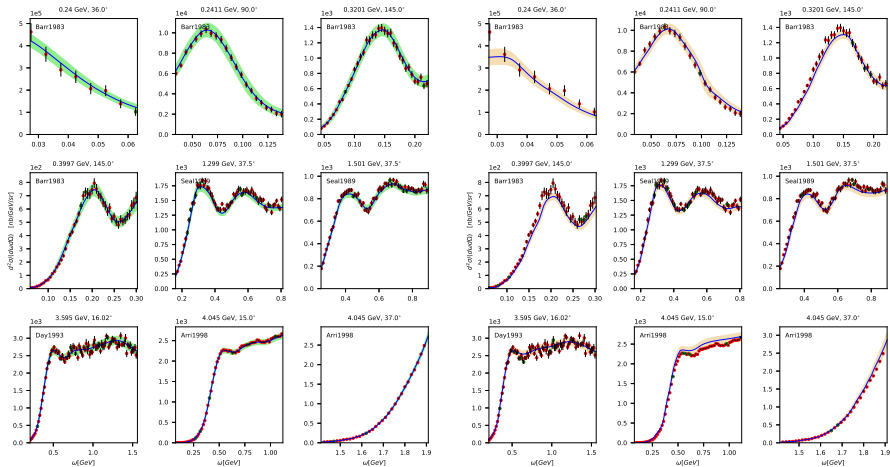
Abbrev.	$\Delta \lambda_k$
Arri1995	4.0%
Arri1998	4.0%
Bagd1988	10.0%
Bara1988	3.7%
Barr1983	2.0%
Dai2018	2.2%
Day1993	3.4%
Fomi2010	4.0%
O'Con1987	5.0%
Seal1989	2.5%
Whit1974	3.0%

- λ_k - normalisation parameters
- $\Delta \lambda_k$ - systematic normalisation error

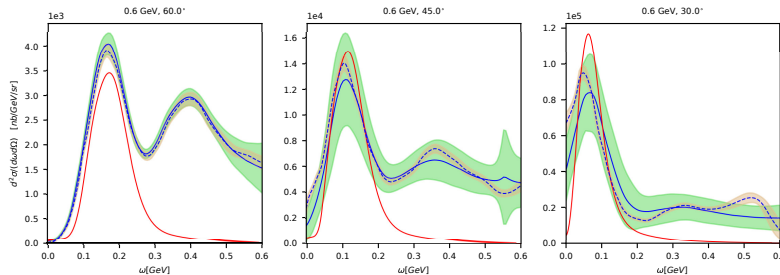
- ▶ Jax package
- ▶ HDF5 to store and organize data
- ▶ AdamW algorithm with decay width 0.004
- ▶ Minibatch configuration with five batches
- ▶ 60 000 epoch
- ▶ We split the dataset into training and test datasets, with a proportion of 9:1



Results: Bootstrap model, Dropout Model



- ▶ training and test points



- ▶ **Model A** Bootstrap and **Model B** MC Dropout ($p=0.01$)
- ▶ **Spectral function** QE scattering, Ankowski, Benhar, Sakuda, PRD 91, (2015) 03300
- ▶ Energy of 600 MeV relevant for neutrino-oscillation experiments such as T2K and the Short Baseline Neutrino program
- ▶ **Bootstrap model generalizes better**

- ▶ we obtained fits to electron-carbon
- ▶ we compared their predictions to
 - ▶ a test dataset,
 - ▶ a theoretical predictions obtained within the spectral function approach
- ▶ Bootstrap model generalizes better than MC Dropout model

In further analysis, we decided to use the bootstrap model

Fits are available from

<https://github.com/bekowal/CarbonElectronNeuralNetwork>

Step 2

Transfer learning

Transferring physics from carbon to other targets

We consider the inclusive electron-nucleus scattering:

$$e + N \rightarrow e + X$$

Transfer learning (TL) is a phenomenon in psychology and education. The ability of a person who has learned skills in one specific field to easily acquire skills needed in related areas of life.

TL can be applied in deep learning:
The DNN learns the details of one kind of data, and after fine-tuning, it makes predictions for related problem. (e.g.: medical images)

We propose to use the TL technique in physics

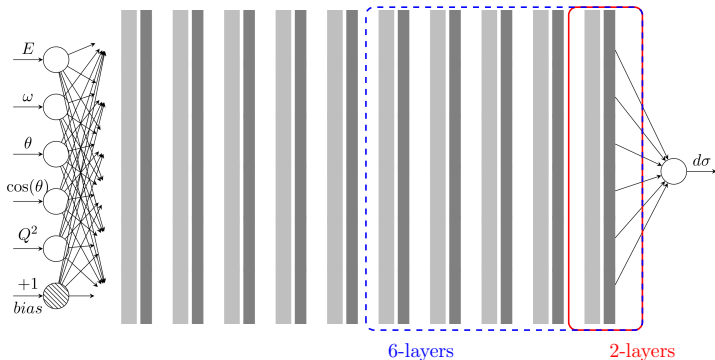
Hypothesis: The network, trained on electron-carbon data, has learned nuclear structure features that are applicable to other nuclei, that share common reaction mechanisms: such as quasielastic scattering, resonance excitation, and deep inelastic scattering

The neurons

- ▶ in the first layers of DNN extract basic features from the input.
- ▶ in the last layers combine these features into more complex ones.

Transfer learning

Beyond the scenario in which we analyze the DNNs in which all layers are fine-tuned, we also consider the situation in which **some number of last layer is fine-tuned**.



the first layers recognizes basic features

subsequent layers combines them into more complex ones

10.1162/neco.2006.18.7.1527, A Fast Learning Algorithm for Deep Belief Nets

If the problems are similar, there is no need to teach multiple layers

Transfer learning

Transfer learning allows a deep neural network trained on one type of data to be adapted for new problems **with limited information**

Training-to-test dataset split:

- ▶ 7:3
- ▶ 1:9

Fine-tuning of DNNs (trained on inclusive electron-carbon) on **other nuclear targets**:

- ▶ helium (3He)
- ▶ lithium (6Li)
- ▶ oxygen (^{16}O)
- ▶ calcium (^{40}Ca),
- ▶ aluminum (^{27}Al),
- ▶ iron (^{56}Fe)

To fine-tune the model - **several hundred epochs**
(in comparison to 60 000 for the base model)

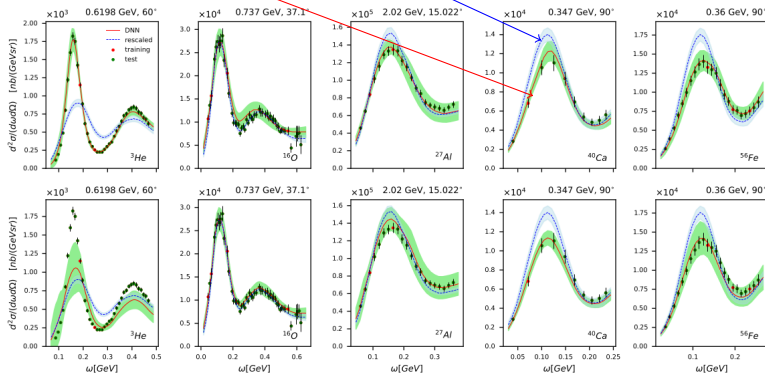
1:9 training-to-test

Double-differential cross section for inclusive electron scattering on He, O, Al, Ca, and Fe for selected kinematics:

base model

model after fine tuning

3

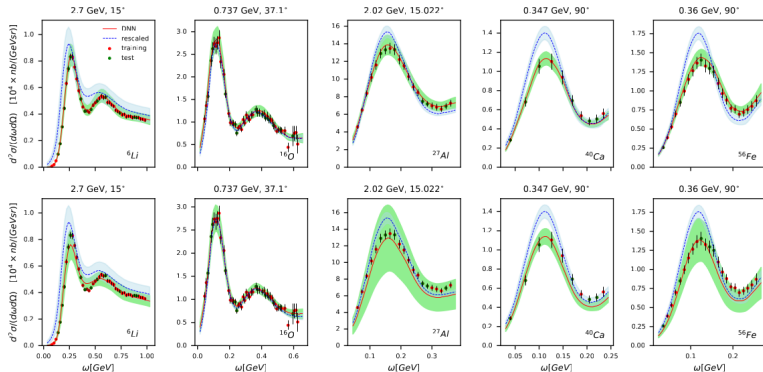


The fits obtained for the proportion of training to test datasets 1 : 9. The results for DNN with all/two last layers fine-tuned are shown in the top/bottom row.

► training and test points

Double-differential cross section for inclusive electron scattering on Li, O, Al, Ca, and Fe for selected kinematics:

2

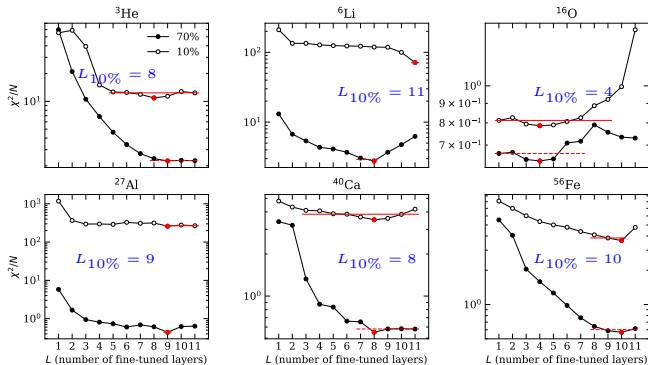


The fits obtained for the proportion of training to test datasets 7 : 3. The results for DNN with all/two last layers fine-tuned are shown in the top/bottom row.

► training and test points

- the full fine-tuned model works better - the distribution of the nuclear structure information in the network is nontrivial.
- both fine-tuning scenario models work well for O, Ca, and Fe
- for Li, Al, He the full fine-tuning gives significantly more accurate model than the shallow fine-tuned one.

χ^2/N_T layer-wise fine tuning



χ^2/N_T , normalized by the number of test points N_T , computed on the test dataset as a function of the number of tuned layers

a reference horizontal plateau marker - mediana between the best and the worst in the region 20% above minimum

The relationship between the carbon and target kinematic domains:

- ▶ **A:** The kinematic domain is covered by the carbon data. Only a few target-domain data points are used for training.
- ▶ **B:** The kinematic domain is covered by the carbon data. No data points from this part of the target domain are used in training.
- ▶ **C:** The kinematic domain is not covered by the carbon data.

A: The kinematic domain is covered by the carbon data. Only a few target-domain data points

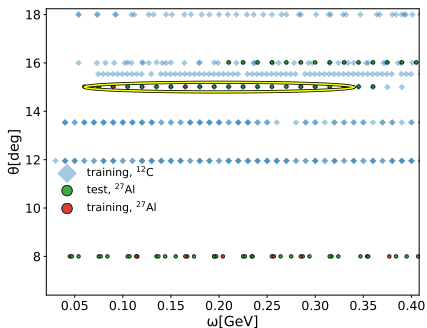


Figure: Carbon data utilized for training the baseline model (blue points) and the aluminum data used to fine-tune the aluminum model.

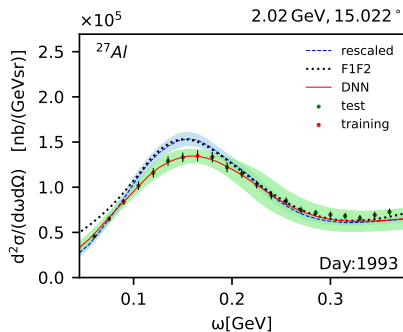


Figure: Case A: aluminum cross sections predicted by 10% training split model with 9 layers tuned. Electron energy 2.02GeV and scattering angle $\theta = 15.022^\circ$

- training, test points, Carbon points
- Rescaled base model
- F1F2 based on the superscaling model

A: The kinematic domain is covered by the carbon data. Only a few target-domain data points

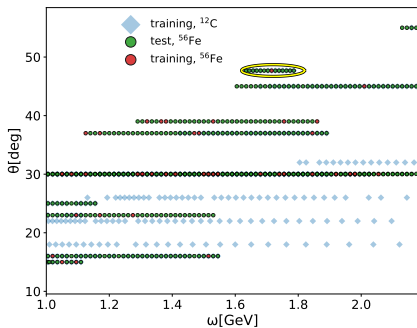


Figure: Carbon data utilized for training the carbon baseline model, the iron data used for tuning the model (red points)

- training, test points, Carbon points
- Rescaled base model
- F1F2 based on the superscaling model

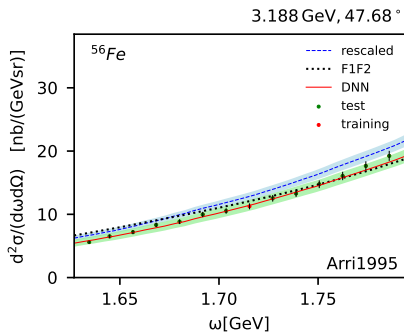


Figure: Case A: iron cross sections predicted by 10% training split model with 10 layers tuned. Electron energy 3.188 GeV and scattering angle $\theta = 47.68^\circ$.

B: The kinematic domain is covered by the carbon data. No target-domain data points

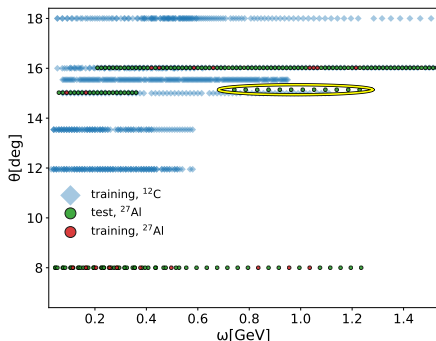


Figure: Case B: carbon data utilized for training the baseline model and the aluminum data used to fine-tune the aluminum model.

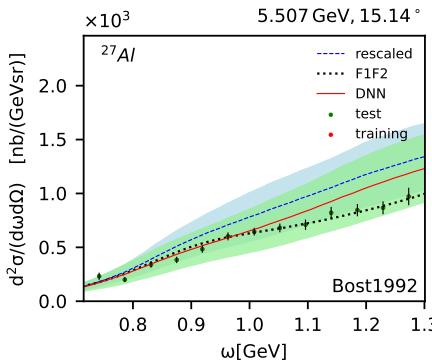


Figure: Case B: aluminum cross sections predicted by 10% training split model with 9 layers tuned. Electron energy 5.507 GeV and scattering angle $\theta = 15.14^\circ$.

- ▶ training, test points, Carbon points
- ▶ Rescaled base model

C: The kinematic domain is not covered by the carbon data.

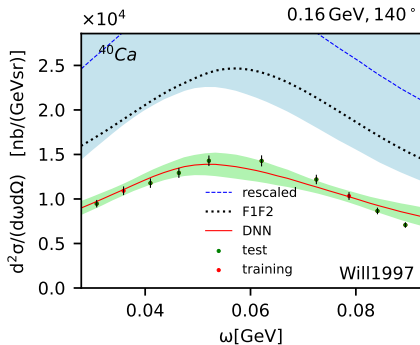


Figure: Case C: calcium cross sections predicted by 10% training split model with 8 layers tuned. In both panels, electron energies 0.16 GeV and scattering angles $\theta = 140^\circ$.

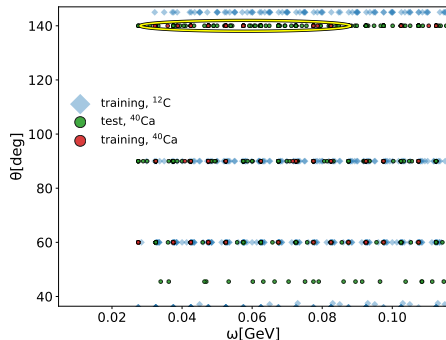


Figure: Carbon data utilized for training the baseline model and the calcium data used to fine-tune the model.

- ▶ training, test points, Carbon points
- ▶ Rescaled base model
- ▶ F1F2 based on the superscaling model

C: The kinematic domain is not covered by the carbon data.

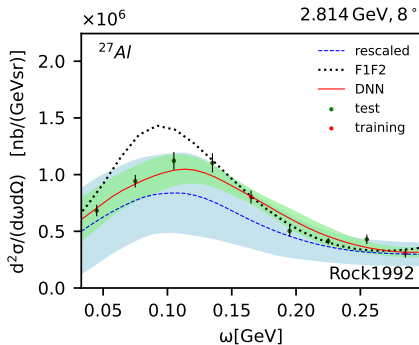


Figure: Case C: aluminum cross sections predicted by 10% training split model with 9 layers tuned. Electron energy 2.814 GeV and scattering angle $\theta = 8^\circ$.

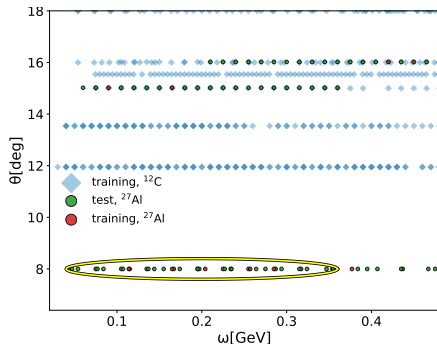


Figure: Case C: carbon data utilized for training the baseline model and the aluminum data used to fine-tune the aluminum model.

- ▶ training, test points, Carbon points
- ▶ Rescaled base model
- ▶ F1F2 based on the superscaling model

C: The kinematic domain is not covered by the carbon data.

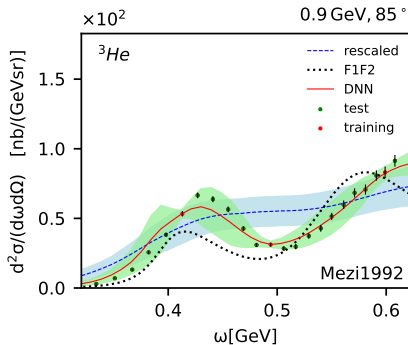


Figure: Case C: helium cross sections predicted by 10% training split model with 8 layers tuned. In both panels, electron energies 0.9 GeV and scattering angles $\theta = 85^\circ$.

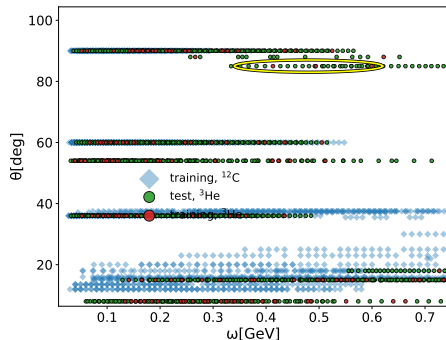


Figure: Carbon data utilized for training the baseline model and the helium data used to fine-tune the model.

- ▶ training, test points, Carbon points
- ▶ Rescaled base model
- ▶ F1F2 based on the superscaling model

The obtained model interpolates and extrapolates well for all three cases

^3He , despite its few-body nature, still admits a quite successful adaptation

^6Li represents the weakest case, reflecting both the limited available dataset

^{27}Al demonstrates that the transfer-learning quality can depend sensitively on a small subset of highly constraining measurements

- ▶ DNNs encode fundamental properties of the nuclear medium, which can be leveraged through the TL techniques to model related processes
- ▶ TL works even with limited information.
- ▶ In particular, our proposed technique holds great promise to be applied to additional targets and to model neutrino-nucleus scattering.

Thank you